

TOLL MANAGEMENT SYSTEM

Project Report

Technologies Used

Python | MySQL

Developed By

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INTRODUCTION

This project is about a Toll management this system calculates toll costs based on vehicle type, distance travelled and the specific toll plaza, ensuring accurate charges fastag accounts can be easily recharged online through credit card, debit card.

MODULES/PACKAGES USED

- `mysql.connector` – To connect with mysql with python.
- `tabulate`—To display the data in a table in tabular form.
- `time`—To get current time.
- `datetime`—To get current date and time.
- `random`-to generate random number

USER DEFINED FUNCTIONS USED

- **sign_in():**To create new account
- **log_in():**To use the account
- **username_password():**To create username and password
- **forgot_password():**To change the password
- **view_account():**To view toll account details
- **delete_account():**To delete toll account
- **update_account():**To update toll account
- **toll_options():**Services provided for the user

- `single_journey()`: Calculating toll cost for single journey
- `round_journey()`: Calculating toll cost for round journey
- `monthly_pass()`: Calculating toll cost for monthly
- `toll_estimation()`: Selecting the destination
- `notify_user()`: checking low balance
- `check_balance()`: checking fastag account balance
- `fastag_registration()`: To create fastag account
- `fastag_options()`: Services provided to user for fastag account users
- `view_record()`: To view record of fastag account

- `update_record()`:To update record of fastag account
- `delete_record()`:To delete the fastag account
- `fastagaccount_details()`:To log into the account
- `credit_card()`:payement purpose
- `debit_card()`:payement purpose
- `recharge_options()`: To recharge fastag account
- `menu_options()`:Home page
- `exit()`:To exit from the program

SOURCE CODE

```
import tabulate
import random
from datetime import date
import time
import mysql.connector as ms
x=ms.connect(host='localhost',user='root',passwd='root',database='toll_management')
cur=x.cursor()
cur.execute('use toll_management')
#notification
def notify_user():
    print('Warning:LOW BALANCE')
    print("")
    choice=input('do you want to recharge YES/NO')
    if choice=='yes':
        fastag_account()
def check_balance(rfid_no):
    cur.execute('select account_balance from fastag_recharge_account where rfid_no="{}".format(rfid_no)')
    data=cur.fetchone()
    balance=data[0]
```

```

def checking():
    low_limit=100
    if balance>low_limit:
        pass
    else:
        notify_user()

def username_password():
    username=input('Create a USERNAME')
    ans='y'
    while ans=='y':
        password=input('Enter the strong password')
        if len(password) < 8:
            print("Password must contain at least 8 characters.")
            ans='y'
        else:
            ans='n'

    cur.execute('insert into username_password (username,password)
values("{}","{}".format(username,password))

    x.commit()

    log_in()

def sign_in():

print('-----')
print('SIGN_IN')

```

```
print('-----')
print("")
name=input('Enter your NAME')
print("")
contact_no=int(input('Enter your PHNO_NO'))
print("")
email_id=input('Enter your EMAIL_ID')
print("")
vehicle_no=input('Enter your VEHICLE_NO')
print("")
vehicle_type=input('Enter the VEHICLE_TYPE')
print("")
vehicle_brand=input('Enter your VEHICLE_BRAND')
print("")
ans='y'
while ans=='y':
    captcha=random.randrange(1000,9999)
    print('captcha:',captcha)
    code=int(input('Enter the above captcha'))
    if captcha==code:
        ans='n'
    else:
        print()
```

```

        print('INCORRECT')

    print()

    print('Enter again')

check=str(contact_no)

user_id=name[0:2].upper()+check[6:12]

print("")

print('Your user_id is',user_id)

print("")

    cur.execute("insert into user_detail
values('{}',{},{},{},{},{},{},{})".format(name,contact_no,email_id,vehicle_n
o,vehicle_type,vehicle_brand,user_id))

    x.commit()

    print()

    print('ACCOUNT CREATED SUCCESSFULLY')

    print("")

    username_password()

def forgot_password():

    user_id=input('enter your user_id')

    cur.execute('select exists(select name from user_detail where
user_id="{}")'.format(user_id))

    data=cur.fetchall()

    if data[0][0]==1:

        ans='y'

        while ans=='y':

            password=input('enter the new password')

```

```

        check=input('enter again the new password')

        if check==password:

            cur.execute('update username_password set password="{}" where
user_id={} '.format(password,user_id))

            print('password changed successfully')

            ans='n'

        else:

            ans='y'

    menu_options()

def view_account():

    print("")

    print('-----')
    print('*****TOLL MANAGEMENT
SYSTEM*****')

    print('-----')

    print("")

    user_id=input('enter the user_id')

    print("")

    print('Enter any one of these data:')

    print("")

    print('1 ⇐ NAME')

    print("")

```

```

print('2⇐CONTACT NO')

print("")

print('3⇐EMAIL_ID')

print("")

print('4⇐VEHICLE_NO')

print("")

choice=int(input('enter your choice'))

print("")

if choice==1:

    name=input('enter the name')

    ans='y'

    while ans=='y':

        if name=="":

            print('MANDATORY')

            ans='y'

        else:

            cur.execute('select *from user_detail where name="{}" and

user_id="{}".format(name,user_id))

            data=cur.fetchall()

h=['NAME','CONTACT_NO','EMAIL_ID','VEHICLE_NO','VEHICLE_CAR','

VEHICLE_BRAND']

            print(tabulate.tabulate(data,headers=h,tablefmt='psql'))

            ans='n'

    elif choice==2:

```

```

contact_no=int(input('enter the contact_no'))

ans='y'

while ans=='y':

    if contact_no=="":

        print('MANDATORY')

        ans='y'

    else:

        cur.execute('select *from user_detail where contact_no={} and
user_id=" {}".format(contact_no,user_id))

        data=cur.fetchall()

h=['NAME','CONTACT_NO','EMAIL_ID','VEHICLE_NO','VEHICLE_CAR','
VEHICLE_BRAND']

print(tabulate.tabulate(data,headers=h,tablefmt='psql'))

elif choice==3:

    email_id=input('enter the email_id:')

    ans='y'

    while ans=='y':

        if email_id=="":

            print('MANDATORY')

            ans='y'

        else:

            cur.execute('select *from user_detail where email_id="{}" and
user_id=" {}".format(email_id,user_id))

            data=cur.fetchall()

```

```
h=['NAME','CONTACT_NO','EMAIL_ID','VEHICLE_NO','VEHICLE_CAR','  
VEHICLE_BRAND']
```

```
print(tabulate.tabulate(data,headers=h,tablefmt='psql'))
```

```
elif choice==4:
```

```
vehicle_no=input('Enter your VEHICLE_NO')
```

```
ans='y'
```

```
while ans=='y':
```

```
    if vehicle_no=="":
```

```
        print('MANDATORY')
```

```
        ans='y'
```

```
    else:
```

```
        cur.execute('select *from user_detail where vehicle_no="{}" and  
user_id="{}"'.format(vehicle_no,user_id))
```

```
        data=cur.fetchall()
```

```
h=['NAME','CONTACT_NO','EMAIL_ID','VEHICLE_NO','VEHICLE_CAR','  
VEHICLE_BRAND']
```

```
print(tabulate.tabulate(data,headers=h,tablefmt='psql'))
```

```
else:
```

```
    print("")
```

```
    print('INVALID INPUT')
```

```
    print("")
```

```
    print("")
```

```
def delete_account():
```

```
    user_id=input('enter the user_id')
```

```

ans='y'
while ans=='y':
    print('select any of the data to delete account:')
    print("")
    print('1 ⇄ Email Id')
    print("")
    print('2 ⇄ Contact no')
    print("")
    print('3 ⇄ Vehicle number')
    print("")
    print('4 ⇄ vehicle type')
    print("")
    print('5 ⇄ vehicle brand')
    print("")
    choice=int(input('Enter your choice'))
    if choice==1:
        email_id=input('enter your email_id to be deleted')
        ans='y'
        while ans=='y':
            if email_id=="":
                print('MANDATORY')
                ans='y'
            else:
                cur.execute('delete from user_detail where emailid="{}" and
user_id="{}".format(email_id,user_id))

```

```

elif choice==2:

    contact_no=int(input('enter your contact_no to be deleted'))

    ans='y'

    while ans=='y':

        if contact_no=="":

            print('MANDATORY')

            ans='y'

        else:

            cur.execute('delete from user_detail where contact_no={} and
user_id={} {}'.format(contact_no,user_id))

    elif choice==3:

        v_no=input('enter your vehicle_no to be deleted')

        ans='y'

        while ans=='y':

            if v_no=="":

                print('MANDATORY')

                ans='y'

            else:

                cur.execute('delete from user_detail where vehicle_no={} and
user_id={} {}'.format(v_no,user_id))

    elif choice==4:

        v_type=input('enter your vehicle_type to be deleted')

        ans='y'

        while ans=='y':

            if name=="":

```

```

        print('MANDATORY')

        ans='y'

    else:

        cur.execute('delete from user_detail where vehicle_type="{}" and
user_id="{}"'.format(v_type,user_id))

    elif choice==5:

        v_brand=input('enter your vehicle_brand to be deleted')

        ans='y'

        while ans=='y':

            if v_brand=="":

                print('MANDATORY')

                ans='y'

            else:

                cur.execute('delete from user_detail where vehicle_brand="{}" and
user_id="{}"'.format(v_brand,user_id))

        else:

            print("")

            print('INVALID INPUT')

            print("")

            x.commit()

            print("")

            print('-----DELETED
SUCESSFULLY-----')

            print("")

            ans=input('Do you want to delete more records')

```

```
    print("")
def update_account():

    user_id=input('enter the user_id')
    ans='yes'
    while ans.lower()=='yes':
        print("")
        print('select the any one of these which one to update the account:')
        print("")
        print('1 ⇔ Email Id')
        print("")
        print('2 ⇔ Contact no')
        print("")
        print('3 ⇔ Vehicle number')
        print("")
        print('4 ⇔ vehicle type')
        print("")
        print('5 ⇔ vehicle brand')
        print("")
        ans='y'
        while ans=='y':
            if user_id=="":
                print('MANDATORY')
                ans='y'
```

```

else:
    ans='n'
print("")
choice=int(input('enter the choice'))
print("")
if choice==1:
    email_id=input('enter your new email_id')
    ans='y'
    while ans=='y':
        if email_id=="":
            print('MANDATORY')
            ans='y'
        else:
            cur.execute('update user_detail set email_id="{}" where
user_id="{}"'.format(email_id,user_id))
            ans='n'
    elif choice==2:
        contact_no=int(input('enter your new contact_no'))
        ans='y'
        while ans=='y':
            if contact_no=="":
                print('MANDATORY')
                ans='y'
            else:

```

```
        cur.execute('update user_detail set contact_no={} where  
user_id={} {}'.format(contact_no,user_id))
```

```
        ans='n'
```

```
    elif choice==3:
```

```
        v_no=input('enter your new vehicle_no ')
```

```
        ans='y'
```

```
        while ans=='y':
```

```
            if v_no=="":
```

```
                print('MANDATORY')
```

```
                ans='y'
```

```
            else:
```

```
                cur.execute('update user_detail set vehicle_no={} {} where  
user_id={} {}'.format(v_no,user_id))
```

```
                ans='n'
```

```
    elif choice==4:
```

```
        v_type=input('enter your new vehicle_type')
```

```
        ans='y'
```

```
        while ans=='y':
```

```
            if v_type=="":
```

```
                print('MANDATORY')
```

```
                ans='y'
```

```
            else:
```

```
                cur.execute('update user_detail set vehicle_type={} {} where  
user_id={} {}'.format(v_type,user_id))
```

```
                ans='n'
```

```

elif choice==5:

    v_brand=input('enter your new vehicle_brand')

    ans='y'

    while ans=='y':

        if v_brand=="":

            print('MANDATORY')

            ans='y'

        else:

            cur.execute('update user_detail set vehicle_brand="{}" where
user_id="{}"'.format(v_brand,user_id))

            ans='n'

        else:

            print('INVALID INPUT')

            x.commit()

            print("")

            print("-----Updated
successfully-----")

            print("")

            ans=input('Do you want to update more records YES/NO')

            print("")

def exit_():

    ans='n'

    print("")

    print("-----LOGGED
OUT-----")

```

```

    print("")
def toll_options():
    print("")

    print('-----')
    print('*****TOLL MANAGEMENT
SYSTEM*****')

    print('-----')
    print("")
    ans='y'
    while ans=='y':
        print("")
        print("    ⇨ 1.View record    ")
        print("")
        print("    ⇨ 2.Update record  ")
        print("")
        print("    ⇨ 3.Delete record  ")
        print("")
        print("    ⇨ 4.Toll estimation ")
        print("")
        print("    ⇨ 5.Fastag registration ")
        print("")
        print("    ⇨ 6.Fastag account   ")

```

```

print("")
print("                ⇨ 7.Exit                ")
print("")
choice=int(input('enter your choice'))
print("")
if choice==1:
    view_account()
elif choice==2:
    update_account()
elif choice==3:
    delete_account()
elif choice==4:
    toll_estimation()
elif choice==5:
    fastag_registration()
elif choice==6:
    fastagaccount_details()
elif choice==7:
    exit_()
    break
else:
    print('Invalid choice')
def log_in():
    print("")

```

```

print("")

print('-----')
print('LOG_IN')

print('-----')

print("")
ans='y'
while ans=='y':
    print("")
    username=input('Enter the username:')
    print("")
    cur.execute("select exists(select password from username_password where
username='{ }')".format(username))
    data=cur.fetchall()
    if data[0][0]!=1:
        print('Username does not exist')
    else:
        ans='n'
ans='y'
while ans=='y':
    print("")
    password=input('enter the password')
    print("")

```

```

cur.execute('select password from username_password where
username="{}".format(username))

data=cur.fetchall()

if password==data[0][0]:

    ans='n'

else:

    print('INCORRECT PASSWORD')

    choice=input('Forgot password YES/NO')

    if choice.lower()=='yes':

        forgot_password()

    else:

        ans='y'

x.commit()

toll_options()

def single_journey(start_end):

    cur.execute('select place from start_destination where
sno={}'.format(start_end))

    data=cur.fetchall()

    place=data[0][0]

    pl=place.replace('-', '_')

    print("")

    cvrwd=input('commercial vehicle registered within district YES/NO')

    print("")

    if cvrwd=='yes'.lower():

        v_type=input('enter the vehicle_type')

```

```

cur.execute('select cvrwd from cost_details where
type_vehicle={} {}'.format(v_type))

cost=cur.fetchone()

cur.execute('select SNO from {}'.format(pl))

data=cur.fetchall()

print(data)

for i in data:

    count=i[0]

cur.execute('select*from {}'.format(pl))

data=cur.fetchall()

print("")

print("")

print('-----')
print('-----')

print('                ',place,'                ')

print('-----')
print('-----')

h=['SNO','TOLL_PLAZA_NAME','DISTANCE','STATE','DISTRICT']

print(tabulate.tabulate(data,headers=h,tablefmt='fancy_grid'))

total=cost[0]*count

print("")

print("                TOTAL COST=",total)

ans=input('do you want to see more places YES/NO:')

print("")

```

```

if ans.lower()=='yes':
    toll_estimation()
    print("")
else:
    v_type=input('enter the vehicle_type')
    cur.execute('select single_journey from cost_details where
type_vehicle=" {}".format(v_type))
    cost=cur.fetchone()
    cur.execute('select SNO from {}'.format(pl))
    data=cur.fetchall()

    for i in data:
        count=i[0]
        cur.execute('select*from {}'.format(pl))
        data=cur.fetchall()
        print('-----',place,'-----')
        h=['SNO','TOLL_PLAZA_NAME','DISTANCE','STATE','DISTRICT']
        print(tabulate.tabulate(data,headers=h,tablefmt='fancy_grid'))
        total=cost[0]*count
        print("")
        print("                                TOTAL COST=",total)
        print("")
        ans=input('do you want to see more places YES/NO:')
        print("")
        if ans.lower()=='yes':

```

```

        toll_estimation()

    print("")

def round_journey(start_end):

    cur.execute('select place from start_destination where
sno={}'.format(start_end))

    data=cur.fetchall()

    place=data[0][0]

    pl=place.replace('-', '_')

    print("")

    v_type=input('enter the vehicle_type')

    print("")

    cur.execute('select return_journey from cost_details where
type_vehicle="{}".format(v_type))

    cost=cur.fetchone()

    cur.execute('select SNO from {}'.format(pl))

    data=cur.fetchall()

    for i in data:

        count=i[0]

    cur.execute('select*from {}'.format(pl))

    data=cur.fetchall()

    print('-----',place,'-----')

    h=['SNO','TOLL_PLAZA_NAME','DISTANCE','STATE','DISTRICT']

    print(tabulate.tabulate(data,headers=h,tablefmt='fancy_grid'))

    total=cost[0]*count

    print("                                TOTAL COST=",total)

```

```

print("")
ans=input('do you want to see more places YES/NO:')
print("")
if ans.lower()=='yes':
    toll_estimation()
    print("")
def monthly_pass(start_end):
    cur.execute('select place from start_destination where
sno={}'.format(start_end))
    data=cur.fetchall()
    place=data[0][0]
    pl=place.replace('-', '_')
    print("")
    v_type=input('enter the vehicle_type')
    print("")
    cur.execute('select monthly_pass from cost_details where
type_vehicle="{}".format(v_type))
    cost=cur.fetchone()
    cur.execute('select SNO from {}'.format(pl))
    data=cur.fetchall()
    for i in data:
        count=i[0]
    cur.execute('select*from {}'.format(pl))
    data=cur.fetchall()
    print('-----',place,'-----')

```

```

h=['SNO','TOLL_PLAZA_NAME','DISTANCE','STATE','DISTRICT']

print(tabulate.tabulate(data,headers=h,tablefmt='fancy_grid'))

total=cost[0]*count

print("                                TOTAL COST=",total)

print("")

ans=input('do you want to see more places YES/NO:')

print("")

if ans.lower()=='yes':

    toll_estimation()

    print("")

def toll_estimation():

    cur.execute('select*from start_destination')

    data=cur.fetchall()

    newdata=[]

    for i in data:

        newdata.append([i[0],i[1]])

    print("")

print('-----')
print('-----')

print("                                TOLL PLAZA LIST                                ")

print('-----')
print('-----')

print("")

h=['SNO','TOLL_PLAZA_NAME']

```

```

print("")
information=tabulate.tabulate(newdata,headers=h,tablefmt='fancy_grid')
print(information)
print("")
ans='y'
while ans=='y':
    start_end=input('enter the sno start_end')
    cur.execute('select exists(select place from start_destination where
sno={})'.format(start_end))
    data=cur.fetchall()
    if data[0][0]==1:
        ans='n'
    else:
        print('invalid input')
        ans='y'
    cur.execute('select place from start_destination where
sno={}'.format(start_end))
    data=cur.fetchall()
    place=data[0][0]
    pl=place.replace('-', '_')
    cur.execute('select*from cost_details')
    data=cur.fetchall()
    print("")
    print("")
h=['Type_vehicle','Single_Journey','Return journey','Monthly pass','Cvrwd']

```

```

print(tabulate.tabulate(data,headers=h,tablefmt='fancy_grid'))

print("")

print("")

print('-----TYPES OF
JOURNEY-----')

print("")

print("")

print('1.Single journey')

print("")

print('2.Round journey')

print("")

print('3.Monthly pass')

print("")

ans='y'

while ans=='y':

    print("")

    choice=int(input('enter the type of journey'))

    print("")

    print("")

    if choice==1:

        single_journey(start_end)

        ans='n'

    elif choice==2:

        round_journey(start_end)

        ans='n'

```

```
    elif choice==3:
        monthly_pass(start_end)
        ans='n'
    else:
        print('invalid input')
def fastag_options():
    print("")
    print('1.Recharge_options')
    print("")
    print('2.view account')
    print("")
    print('3.update account')
    print("")
    print('4.delete account')
    print("")
    print('5.exit')
    print("")
    choice=int(input('enter your choice'))
    ans='y'
    while ans=='y':
        if choice==1:
            recharge_options()
            ans='n'
        elif choice==2:
```

```

        view_record()

        ans='n'

    elif choice==3:

        update_record()

        ans='n'

    elif choice==4:

        delete_record()

        ans='n'

    elif choice==5:

        toll_options()

    else:

        print('invalid input')

def fastag_registration():

print('-----')

print('                FASTAG REGISTRATION

')

print('-----')

print("")

name=input('Enter your NAME')

print("")

contact_no=int(input('Enter your PHNO_NO'))

print("")

```

```
email_id=input('Enter your EMAIL_ID')
print("")
vehicle_no=input('Enter your VEHICLE_NO')
print("")
vehicle_type=input('Enter the VEHICLE_TYPE')
print("")
vehicle_brand=input('Enter your VEHICLE_BRAND')
print("")
ans='y'
while ans=='y':
    print("")
    aadhar_card=input('Enter your aadhar card')
    print("")
    if len(aadhar_card)==14:
        ans='n'
    else:
        print("")
        print('INVALID AADHAR NUMBER')
        print("")
ans='y'
while ans=='y':
    print("")
    pan_no=input('Enter the Pan card number')
    print("")
```

```
check1=pan_no[0:4]
check2=pan_no[-1]
if check1.isalpha():
    if check2.isalpha():
        ans='n'
    else:
        print("")
        print('INVALID PAN CARD NUMBER')
        print("")
        ans='y'
ans='y'
while ans=='y':
    print("")
    passport_no=input('Enter your passport number')
    print("")
    if len(passport_no)==8:
        ans='n'
    else:
        print("")
        print('INVALID PASSPORT NUMBER')
ans='y'
while ans=='y':
    captcha=random.randrange(1000,9999)
    print(captcha)
```

```

code=int(input('Enter the above captcha'))

if captcha==code:
    ans='n'
else:
    print()
    print('INCORRECT')
    print()
    print('Enter again')

ans='y'

while ans=='y':
    rfid_no=random.randrange(1000,9999)

    cur.execute("select exists(select name from fastag_accountdetails where
rfid_no='{ }')".format(rfid_no))

    data=cur.fetchall()

    if data[0][0]!=1:
        ans='n'
    else:
        ans='y'

    cur.execute("insert into fastag_accountdetails
values('{ }',{ },'{ }',{ }',{ }',{ }',{ }',{ }',{ }',{ }')".format(name,contact_no,email_id,vehic
hicle_no,vehicle_type,aadhar_card,pan_no,passport_no,rfid_no))

    x.commit()

#create username_password for fastag_account

username=input('Create a USERNAME')

ans='y'

```

```

while ans=='y':
    password=input('Enter the strong password')
    if len(password) < 8:
        print("Password must contain at least 8 characters.")
        ans='y'
    else:
        ans='n'

    cur.execute('insert into fastag_recharge_account (username,password,rfid_no)
values("{}","{}",{})'.format(username,password,name))

    x.commit()

    print('ACCOUNT CREATED SUCESSFULLY')

    fastag_options()

def view_record():
    name=input('enter the name')
    rfid_no=int(input('enter the rfid_no'))

    cur.execute('select exists(select Vehicle_no from fastag_accountdetails where
rfid_no={} and name=" {}".format(rfid_no,name))

    data=cur.fetchall()

    if data[0][0]==1:
        cur.execute('select *from fastag_accountdetails where
rfid_no={}'.format(rfid_no))

        data=cur.fetchall()

h=['Name','Contact_No','Email_id','Vehicle_No','Vehicle_type','Aadhar_card','Pa
n_card','Passport_no','RFID_No']

    print(tabulate.tabulate(data,headers=h,tablefmt='psql'))

```

```
else:

    print('rfid_no does not exists')

fastag_options()

def update_record():

    ans='y'

    while ans=='y':

        print('select the any one of these to update the account:')

        print("")

        print('1.Email Id')

        print("")

        print('2.Contact no')

        print("")

        print('3.Vehicle number')

        print("")

        print('4.vehicle type')

        print("")

        print('5.vehicle brand')

        print("")

        print('6.Aadhar card')

        print("")

        print('7.Pan card')

        print("")

        print('8.Passport')

        print("")
```

```

rfid_no=int(input('enter your rfid_no '))

print("")

choice=int(input('enter the choice'))

if choice==1:

    email_id=input('enter your new email_id')

    cur.execute('update fastag_accountdetails set email_id="{}" where
rfid_no="{}".format(email_id,rfid_no))

    elif choice==2:

        contact_no=int(input('enter your new contact_no'))

        cur.execute('update fastag_accountdetails set contact_no={} where
rfid_no="{}".format(contact_no,rfid_no))

    elif choice==3:

        v_no=input('enter your new vehicle_number')

        cur.execute('update fastag_accountdetails set vehicle_no="{}" where
rfid_no="{}".format(v_no,rfid_no))

    elif choice==4:

        v_type=input('enter your new vehicle_type')

        cur.execute('update fastag_accountdetails set vehicle_type="{}" where
rfid_no="{}".format(v_type,rfid_no))

    elif choice==5:

        v_brand=input('enter your new vehicle_brand')

        cur.execute('update fastag_accountdetails set vehicle_brand="{}" where
rfid_no="{}".format(v_brand,rfid_no))

    elif choice==5:

        v_brand=input('enter your new aadhar_card number')

        cur.execute('update fastag_accountdetails set aadharcard="{}" where
rfid_no="{}".format(aadhar_no,rfid_no))

```

```

elif choice==5:

    v_brand=input('enter your new pan_card number')

    cur.execute('update fastag_accountdetails set pancard="{}" where
rfid_no="{}"'.format(pan_card,rfid_no))

    elif choice==5:

        v_brand=input('enter your new passport_no')

        cur.execute('update fastag_accountdetails set passport_no="{}" where
rfid_no="{}"'.format(passport_no,rfid_no))

    else:

        print('INVALID INPUT')

        print('Updated successfully')

        ans=input('Do you want to update more columns y/n')

        fastag_options()

def delete_record():

    ans='y'

    while ans=='y':

        print('select the column you need to delete:')

        print('1.Email Id')

        print('2.Contact no')

        print('3.Vehicle number')

        print('4.vehicle type')

        print('5.aadhar_card')

        print('6.pan_card')

        print('7.passport')

        if choice==1:

```

```
email_id=input('enter your email_id to be deleted')

cur.execute('delete from fastag_accountdetails where
emailid="{}".format(email_id))

elif choice==2:

    contact_no=int(input('enter your contact_no to be deleted'))

    cur.execute('delete from fastag_accountdetails where
contact_no={}'.format(contact_no))

elif choice==3:

    v_no=input('enter your vehicle_no to be deleted')

    cur.execute('delete from fastag_accountdetails where
vehicle_no="{}".format(v_no))

elif choice==4:

    v_type=input('enter your vehicle_type to be deleted')

    cur.execute('delete from fastag_accountdetails where
vehicle_type="{}".format(v_type))

elif choice==5:

    aadhar_card=input('enter your aadhar_card number to be deleted')

    cur.execute('delete from fastag_accountdetails where
aadharcard="{}".format(aadhar_card))

elif choice==6:

    pan_card=input('enter your pan_card number to be deleted')

    cur.execute('delete from fastag_accountdetails where
pancard="{}".format(pan_card))

elif choice==7:

    passport_no=input('enter your passport to be deleted')

    cur.execute('delete from fastag_accountdetails where
passport_no="{}".format(passport_no))
```

```

else:

    print('INVALID INPUT')

    ans=input('Do you want to delete more records')

fastag_options()

def fastagaccount_details():

    ans='y'

    while ans=='y':

        print("")

        name=input('enter your name')

        print("")

        rfid_no=input('enter the RFID TAG')

        print("")

        cur.execute('select exists(select rfid_no from fastag_accountdetails where
name=" {} ").format(name))

        data=cur.fetchall()

        if data[0][0]!=1:

            print('rfid tag does not exist')

        else:

            ans='n'

    check_balance(rfid_no)

    ans='y'

    while ans=='y':

        print("")

```

```

username=input('Enter the username')

print("")

cur.execute("select exists(select username from fastag_recharge_account
where rfid_no={})".format(rfid_no))

data=cur.fetchall()

if data[0][0]!=1:

    print('Username does not exist')

    ans='y'

else:

    break

ans='y'

while ans=='y':

    print("")

    password=input('enter the password')

    print("")

    cur.execute('select password from fastag_recharge_account where
rfid_no=" {} "'.format(rfid_no))

    data=cur.fetchall()

    if password==data[0][0]:

        ans='n'

    else:

        print('INCORRECT PASSWORD')

        choice=input('Forgot password YES/NO')

        if choice.lower()=='yes':

            forgot_password()

```

```

        else:
            ans='y'
        x.commit()
    fastag_options()
def credit_card():
    today_date=date.today()
    today=str(today_date)
    holder_name=input('enter the holder_name')
    ans='y'
    while ans=='y':
        exp_month=int(input("enter the expiry month 'MM'"))
        cur.execute('select Expiry_Month from credit_card where
card_holdername="{}"'.format(holder_name))
        data1=cur.fetchone()
        exp_year=int(input("enter the expiry year 'YY'"))
        cur.execute('select Expiry_year from credit_card where
card_holdername="{}"'.format(holder_name))
        data2=cur.fetchone()
        check1=int(data1[0])>=int(today[5:7])
        if int(data2[0])==int(today[2:4]):
            if check:
                ans='n'
            else:
                print('card expired')
        else:

```

```

        ans='n'

ans='y'

while ans=='y':

    cur.execute('select exists(select *from credit_card where
card_holdername="{ }").format(holder_name))

    data=cur.fetchall()

    if data[0][0]==1:

        ans='y'

        while ans=='y':

            credit_cardno=input('enter the credit_card number')

            cur.execute('select exists(select card_number from credit_card where
card_holdername="{ }").format(holder_name))

            data=cur.fetchall()

            if data[0][0]==1:

                ans='n'

            else:

                ans='y'

ans='y'

while ans=='y':

    cur.execute('select account_balance from credit_card where
card_holdername="{ }".format(holder_name))

    balance=cur.fetchone()

    pin_no=int(input('enter the pin number'))

    cur.execute('select pin_number from credit_card where
card_holdername="{ }".format(holder_name))

    data=cur.fetchone()

```

```

if data[0]==pin_no:

    amount=float(input('enter the amount to be recharged'))

    cur.execute('select account_balance from fastag_recharge_account
where name="{}".format(holder_name))

    new=cur.fetchone()

    add=new[0]+amount

    cur.execute('update fastag_recharge_account set account_balance={}
where name="{}".format(add,holder_name))

    remaining=balance[0]-amount

    cur.execute('update credit_card set account_balance="{}" where
card_holdername="{}".format(remaining,holder_name))

    print('Processing payement')

    time.sleep(6)

    print('Payement sucessfull')

    ans='n'

    x.commit()

else:

    print('enter the correct pin')

    ans='y'

    x.commit

    fastag_options()

def debit_card():

    today_date=date.today()

    today=str(today_date)

    holder_name=input('enter the holder_name')

```

```

ans='y'

while ans=='y':

    exp_month=int(input("enter the expiry month 'MM'"))

    cur.execute('select Expiry_Month from debit_card where
card_holdername="{}"'.format(holder_name))

    data1=cur.fetchone()

    exp_year=int(input("enter the expiry year 'YY'"))

    cur.execute('select Expiry_year from debit_card where
card_holdername="{}"'.format(holder_name))

    data2=cur.fetchone()

    check1=int(data1[0])>=int(today[5:7])

    if int(data2[0])==int(today[2:4]):

        if check:

            ans='n'

        else:

            print('card expired')

    else:

        ans='n'

ans='y'

while ans=='y':

    cur.execute('select exists(select *from debit_card where
card_holdername="{}"'.format(holder_name))

    data=cur.fetchall()

    if data[0][0]==1:

        ans='y'

```

```

while ans=='y':

    credit_cardno=input('enter the debit_card number')

    cur.execute('select exists(select card_number from debit_card where
card_holdername="{}".format(holder_name))

    data=cur.fetchall()

    if data[0][0]==1:

        ans='n'

    else:

        ans='y'

ans='y'

while ans=='y':

    cur.execute('select account_balance from debit_card where
card_holdername="{}".format(holder_name))

    balance=cur.fetchone()

    pin_no=int(input('enter the pin number'))

    cur.execute('select pin_number from debit_card where
card_holdername="{}".format(holder_name))

    data=cur.fetchone()

    if data[0]==pin_no:

        amount=float(input('enter the amount to be recharged'))

        cur.execute('select account_balance from fastag_recharge_account
where name="{}".format(holder_name))

        new=cur.fetchone()

        add=new[0]+amount

        cur.execute('update fastag_recharge_account set account_balance={}
where name="{}".format(add,holder_name))

```

```

    remaining=balance[0]-amount

    cur.execute('update debit_card set account_balance="{}" where
Card_Holdername="{}".format(remaining,holder_name))

    print('Processing payement')

    time.sleep(6)

    x.commit()

    print('Payement sucessfull')

    ans='n'

    x.commit()

    break

else:

    print('enter the correct pin')

    ans='y'

fastag_options()

```

```

def recharge_options():

    ans='y'

    while ans=='y':

        captcha=random.randrange(1000,9999)

        print(captcha)

        code=int(input('Enter the above captcha'))

        if captcha==code:

            ans='n'

        else:

            print()

```

```

        print('INCORRECT')

        print()

        print('Enter again')

    print('-----PAYEMENT METHOD-----')

    print("")

    print('1.Credit card')

    print("")

    print('2.Debit card')

    print("")

    choice=int(input('enter your choice'))

    if choice==1:

        credit_card()

        ans='n'

    elif choice==2:

        debit_card()

        ans='n'

    else:

        print('Invalid choice')


def menu_options():

    print("")

print('-----')

```

```
print('*****TOLL MANAGEMENT  
SYSTEM*****')
```

```
print('-----  
-----')
```

```
print("")
```

```
print("")
```

```
1.SIGN_IN
```

```
2.LOG_IN  ")
```

```
print("")
```

```
choice=int(input('enter the choice'))
```

```
if choice==1:
```

```
    sign_in()
```

```
elif choice==2:
```

```
    log_in()
```

```
else:
```

```
    print('invaid input')
```

```
menu_options()
```

OUTPUT

```
-----  
*****TOLL MANAGEMENT SYSTEM*****  
-----
```

1.SIGN_IN

2.LOG_IN

enter the choice1|

```
-----  
SIGN_IN  
-----
```

Enter your NAMEVelmurugan

Enter your PHNO_NO9944552211

Enter your EMAIL_IDvelreach@gmail.com

Enter your VEHICLE_NOTN89F2344

Enter the VEHICLE_TYPECAR

Enter your VEHICLE_BRANDJimny

captcha: 8851

Enter the above captcha8851

Your user_id is VE2211

ACCOUNT CREATED SUCCESSFULLY

Create a USERNAMEvel@2512

Enter the strong passwordhelpmeout@123

LOG_IN

Enter the username:vel@2512

enter the passwordhelpmeout@123

*****TOLL MANAGEMENT SYSTEM*****

- ➡ 1.View record
- ➡ 2.Update record
- ➡ 3.Delete record
- ➡ 4.Toll estimation
- ➡ 5.Fastag registration
- ➡ 6.Fastag account
- ➡ 7.Exit

enter your choice1

*****TOLL MANAGEMENT SYSTEM*****

enter the user_idVE2211

Enter any one of these data:

1⇨NAME

2⇨CONTACT NO

3⇨EMAIL_ID

4⇨VEHICLE_NO

enter your choice1

enter the nameVelmurugan

	NAME	CONTACT_NO	EMAIL_ID	VEHICLE_NO	VEHICLE_CAR	VE
Velmurugan	9944552211	velreach@gmail.com	TN89F2344	CAR	Jimny	VE

*****TOLL MANAGEMENT SYSTEM*****

- ➡ 1.View record
- ➡ 2.Update record
- ➡ 3.Delete record
- ➡ 4.Toll estimation
- ➡ 5.Fastag registration
- ➡ 6.Fastag account
- ➡ 7.Exit

enter your choice2

enter the user_idVE2211

select the any one of these which one to update the account:

- 1➡Email Id
- 2➡Contact no
- 3➡Vehicle number
- 4➡vehicle tvpe

enter the choice1

enter your new email_idVELreach25@gmail.com

-----Updated successfully-----

Do you want to update more records YES/NOYES

select the any one of these which one to update the account:

- 1➡Email Id
- 2➡Contact no
- 3➡Vehicle number
- 4➡vehicle type
- 5➡vehicle brand

enter the choice2

enter your new contact_no9944225211

-----Updated successfully-----

TOLL PLAZA LIST

SNO	TOLL PLAZA NAME
1	Hyderabad-Chennai
2	Hosur-Chennai
3	Chennai-Kodaikanal
4	Attur-Chennai
5	Bangalore-Nellore
6	Lucknow-Danta
7	haldia-durgapur
8	delhi-haridwar
9	krishnagiri-tiruchendur
10	Ghaziabad-Alwar
11	Coimbatore-Chennai
12	Mumbai-Indore
13	Ghorakpur-Lucknow
14	Jodhpur-Udaipur
15	Uttar Pradesh-Kashmir
16	Rajasthan-Narora
17	Rajkot-Dwarka

18	Tirupur-Bangalore
19	Kanpur-Nainital
20	New Delhi-Bidhuna
21	Nanguneri-Krishnagiri

enter the sno start_end1

Type_vehicle	Single_Journey	Return journey	Monthly pass	Cvrwd
Car	70	105	2350	35
Jeep	70	105	2350	35
Van	70	105	2350	35
LCV	115	170	3795	55
Bus	240	360	7955	120
Truck	240	360	7955	120
Upto_3_Axle_Vehicle	260	390	8680	130
4_to_6_Axle	375	560	12475	185
HCM/EME	375	560	12475	185
7_or_more_Axle	455	685	15190	230

-----TYPES OF JOURNEY-----

1.Single journey

2.Round journey

3.Monthly pass

enter the type of journey1

commercial vehicle registered within district YES/NOyes

enter the vehicle_typecar

Hyderabad-Chennai

SNO	TOLL_PLAZA_NAME	DISTANCE	STATE
1	Pantangi,Vijayawada-Hyderabad Hwy	Telengana	Nalgonda
2	Bhramanapalli Toll Plaza, SH 2	Andhra Pradesh	Kadapa
3	Elchuru Toll Plaza, Addanki-Narasarao Pet Rd	Andhra Pradesh	Palwal
4	Tangatur Toll Plaza, Guntur-Chennai Hwy	Andhra Pradesh	Nellore
5	Musunur Toll Plaza, Guntur-Chennai Hwy	Andhra Pradesh	Nellore
6	Nellore Plaza (Nh- 16), Chennai - Srikakulam Hwy	Andhra Pradesh	Tirupati
7	Budhanam Plaza (Nh- 16), Guntur-Chennai Hwy	Andhra Pradesh	Nellore
8	Sullurpet Plaza (Nh-16), Guntur-Chennai Hwy	Andhra Pradesh	Suryapet
9	Nallur Toll Plaza, Chennai - Srikakulam Hwy	Tamil Nadu	Pudukottai

TOTAL COST= 315.0

do you want to see more places YES/NO:yes

TOLL PLAZA LIST

SNO	TOLL PLAZA NAME
1	Hyderabad-Chennai
2	Hosur-Chennai
3	Chennai-Kodaikanal
4	Attur-Chennai
5	Bangalore-Nellore
6	Lucknow-Danta
7	haldia-durgapur
8	delhi-haridwar
9	krishnagiri-tiruchendur
10	Ghaziabad-Alwar
11	Coimbatore-Chennai
12	Mumbai-Indore
13	Ghorakpur-Lucknow
14	Jodhpur-Udaipur
15	Uttar Pradesh-Kashmir
16	Rajasthan-Narora
17	Rajkot-Dwarka

18	Tirupur-Bangalore
19	Kanpur-Nainital
20	New Delhi-Bidhuna
21	Nanguneri-Krishnagiri

enter the sno start_end2

Type_vehicle	Single_Journey	Return journey	Monthly pass	Cvrwd
Car	70	105	2350	35
Jeep	70	105	2350	35
Van	70	105	2350	35
LCV	115	170	3795	55
Bus	240	360	7955	120
Truck	240	360	7955	120
Upto_3_Axle_Vehicle	260	390	8680	130
4_to_6_Axle	375	560	12475	185
HCM/EME	375	560	12475	185
7_or_more_Axle	455	685	15190	230

-----TYPES OF JOURNEY-----

- 1.Single journey
- 2.Round journey
- 3.Monthly pass

enter the type of journey1

commercial vehicle registered within district YES/NO

enter the vehicle_type

----- Hosur-Chennai -----

SNO	TOLL_PLAZA_NAME	DISTANCE	STATE
1	Krishnagiri Plaza,Chennekothapalli Rd	Tamil Nadu	Ranga Reddy
2	Vaniyambadi Toll Plaza, NH 46	Tamil Nadu	Vellore
3	Pallikonda Toll Plaza, Chennai - Bengaluru Hwy	Tamil Nadu	Vellore
4	Chennasamaduram Toll Plaza, NH 46	Tamil Nadu	Erode
5	Sriperumbadur, Perumalpattu - Kottamedu Rd	Tamil Nadu	Kanchipuram

TOTAL COST= 350.0

do you want to see more places YES/NO:yes

TOLL PLAZA LIST

SNO	TOLL PLAZA NAME
1	Hyderabad-Chennai
2	Hosur-Chennai
3	Chennai-Kodaikanal
4	Attur-Chennai
5	Bangalore-Nellore
6	Lucknow-Danta
7	haldia-durgapur
8	delhi-haridwar
9	krishnagiri-tiruchendur
10	Ghaziabad-Alwar
11	Coimbatore-Chennai
12	Mumbai-Indore
13	Ghorakpur-Lucknow
14	Jodhpur-Udaipur
15	Uttar Pradesh-Kashmir
16	Rajasthan-Narora
17	Rajkot-Dwarka

18	Tirupur-Bangalore
19	Kanpur-Nainital
20	New Delhi-Bidhuna
21	Nanguneri-Krishnagiri

enter the sno start_end18

Type_vehicle	Single_Journey	Return journey	Monthly pass	Cvrwd
Car	70	105	2350	35
Jeep	70	105	2350	35
Van	70	105	2350	35
LCV	115	170	3795	55
Bus	240	360	7955	120
Truck	240	360	7955	120
Upto_3_Axle_Vehicle	260	390	8680	130
4_to_6_Axle	375	560	12475	185
HCM/EME	375	560	12475	185
7_or_more_Axle	455	685	15190	230

1.Single journey

2.Round journey

3.Monthly pass

enter the type of journey2

enter the vehicle_typetruck

----- Tirupur-Bangalore -----

SNO	TOLL_PLAZA_NAME	DISTANCE	STATE
1	Ivcr1 Chengapally Tollways,Coimbatore Rd	Tamil Nadu	Erode
2	L&T_Krishnagiri Thopur Toll Plaza,Hogenakkal Rd	Tamil Nadu	Dharmapuri
3	Krishnagiri Plaza,Chennekothapalli Rd	Tamil Nadu	Krishnagiri
4	Attibelle,Srinagar_Kanyakumari Hwy	Karnataka	Bangalore
5	Electronic_City_Phase 1,Hosur Rd	Karnataka	Bangalore

TOTAL COST= 1800.0

do you want to see more places YES/NO:yes

TOLL PLAZA LIST

SNO	TOLL PLAZA NAME
1	Hyderabad-Chennai
2	Hosur-Chennai
3	Chennai-Kodaikanal
4	Attur-Chennai
5	Bangalore-Nellore
6	Lucknow-Danta
7	haldia-durgapur
8	delhi-haridwar
9	krishnagiri-tiruchendur
10	Ghaziabad-Alwar
11	Coimbatore-Chennai
12	Mumbai-Indore
13	Ghorakpur-Lucknow
14	Jodhpur-Udaipur
15	Uttar Pradesh-Kashmir
16	Rajasthan-Narora
17	Rajkot-Dwarka

18	Tirupur-Bangalore
19	Kanpur-Nainital
20	New Delhi-Bidhuna
21	Nanguneri-Krishnagiri

enter the sno start_end9

Type_vehicle	Single_Journey	Return journey	Monthly pass	Cvrwd
Car	70	105	2350	35
Jeep	70	105	2350	35
Van	70	105	2350	35
LCV	115	170	3795	55
Bus	240	360	7955	120
Truck	240	360	7955	120
Upto_3_Axle_Vehicle	260	390	8680	130
4_to_6_Axle	375	560	12475	185
HCM/EME	375	560	12475	185
7_or_more_Axle	455	685	15190	230

-----TYPES OF JOURNEY-----

1.Single journey

2.Round journey

3.Monthly pass

enter the type of journey3

enter the vehicle_typevan

----- krishnagiri-tiruchendur -----

SNO	TOLL_PLAZA_NAME	DISTANCE	STATE
1	krishnagiri thopur Toll Plaza,Jarugu_Hogenakkal Rd	Tamil Nadu	Dharmapuri
2	Omalur Toll Plaza,Kanyakumari Hwy	Tamil Nadu	Salem
3	Rasampalayam Toll Plaza,NH 44	Tamil Nadu	Namakkal
4	Velanchettiyur ,TNDK expy	Tamil Nadu	Karur
5	Kozhijipatti Plaza,Kanyakumari Hwy	Tamil Nadu	Salem
6	Valayangulam Toll Plaza,Kanyakumari Rd	Tamil Nadu	Tenkasi

TOTAL COST= 14100.0

do you want to see more places YES/NO:no

- ➡ 1.View record
- ➡ 2.Update record
- ➡ 3.Delete record
- ➡ 4.Toll estimation
- ➡ 5.Fastag registration
- ➡ 6.Fastag account
- ➡ 7.Exit

enter your choice5

FASTAG REGISTRATION

Enter your NAMEVelmurugan

Enter your PHNO_NO9944225211

Enter your EMAIL_IDVELreach25@gmail.com

Enter your VEHICLE_NOTN89F2344

Enter the VEHICLE_TYPECAR

Enter your VEHICLE_BRANDJIMNY

Enter your aadhar card8956 4523 6589

Enter the Pan card numberPGHFJ5924D

Enter your passport numberT8954123

6429

Enter the above captcha6429

Create a USERNAMEVEL@avan

Enter the strong passwordhelpmeout@gmail.com

ACCOUNT CREATED SUCESSFULLY

1.Recharge_options

2.view account

3.update account

4.delete account

5.exit

enter your choice2

enter the namevelmurugan

enter the rfid_no1125

```
+-----+-----+-----+-----+-----+-----+
| Name      | Contact_No | Email_id      | Vehicle_No | Vehicle_type | Aadhar_card |
| Pan_card  | Passport_no | RFID_No      |            |              |             |
+-----+-----+-----+-----+-----+-----+
| Velmurugan | 9944225211 | VELreach25@gmail.com | TN89F2344 | CAR          | 8956 4523  |
6589 | PGHFJ5924D | T8954123      | 1125      |              |             |
+-----+-----+-----+-----+-----+-----+
```

1.Recharge_options

2.view account

3.update account

4.delete account

5.exit

Enter the above captcha6615

-----PAYEMENT METHOD-----

1.Credit card

2.Debit card

enter your choice1

```
enter the holder_namepriya
enter the expiry month 'MM'01
enter the expiry year 'YY'35
35
enter the credit_card number2891 7632 1095 7652
enter the pin number8652
enter the amount to be recharged250
Processing payment
Payment sucessfull
```

- ⇨ 1.View record
- ⇨ 2.Update record
- ⇨ 3.Delete record
- ⇨ 4.Toll estimation
- ⇨ 5.Fastag registration
- ⇨ 6.Fastag account
- ⇨ 7.Exit

```
enter your choice7
```

-----LOGGED OUT-----

REFERENCES

- WWW.GOOGLE.COM
- WWW.PYTHON.ORG
- WWW.YOUTUBE.COM